



Université de Sherbrooke

Summer school in biodiversity modelling August 24th, 2026

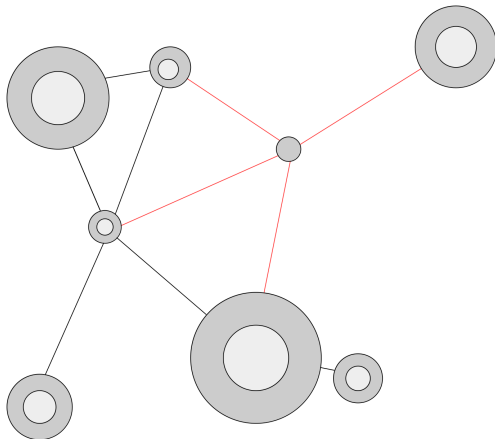
Research projects

Dominique Gravel



Project 1 Metapopulation and climate change

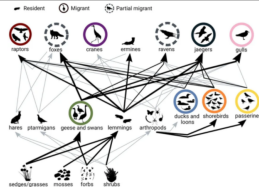
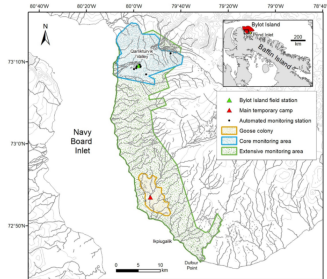
Question : How is climate change impacting metapopulation persistence in mountainous landscapes ?





Project 2 Food web synchrony

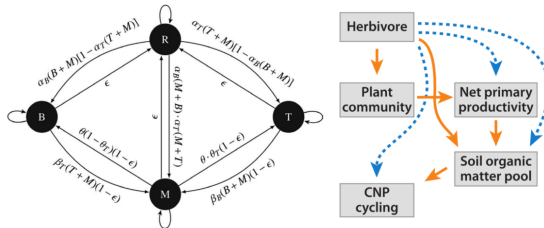
Question : What is the effect of migration on the synchrony of spatial food webs ?





Project 3 Plant-soil feedbacks

Question : How soil-plant interactions impact tree species migration at the temperate-boreal ecotone ?

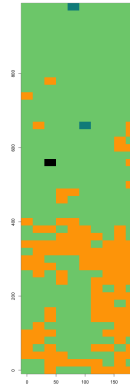




Project 4 Co-occurrence

Question : What is the influence of co-occurrence on beta-diversity ?

- ▶ 200m élévation gradient
- ▶ 500 quadrats
- ▶ 10 m^2 plots
- ▶ Abundance
- ▶ Approx. 100 species
- ▶ 9 years





Project 5 Spatial networks

Question : Are interacting species sharing the same niche ?

- ▶ Host-parasite interactions among salix, leaf galls and paraistoids across Europe;
- ▶ 222 Host species;
- ▶ 146 Parasite species;
- ▶ 4 295 Interactions;
- ▶ 374 Locations;
- ▶ 32 412 species pairs;
- ▶ 27 367 observed co-occurrences;

